

Arithmetic and complexity

Is the matrix multiplication exponent two?

Difficulty: extreme **Importance:** broadly interesting**Status:** Open

Rating rationale: Extreme because closing the exponent gap is a longstanding central barrier in algebraic algorithms; broad importance follows from matrix multiplication's role throughout NLA and computational complexity.

Topic: arithmetic complexity of dense matrix multiplication

Problem statement

Over the field $\mathbb{C}$, let $M(n)$ be the minimum number of scalar additions, subtractions, multiplications, and divisions in an arithmetic straight-line program computing every entry of AB for arbitrary $A, B \in \mathbb{C}^{n \times n}$. Programs must be defined on their intended inputs. Set $\omega = \inf\{\tau : M(n) = O(n^\tau)\}$. Is $\omega = 2$? Equivalently, for every $\varepsilon > 0$, can these products be computed in $O(n^{2+\varepsilon})$ arithmetic operations? This asks about asymptotic arithmetic cost; it does not assert an $O(n^2)$ algorithm or a floating-point stability guarantee.

Why it matters

Matrix multiplication is a basic cost driver for dense NLA.

References and status

M. Bläser, *Fast Matrix Multiplication*, Graduate Surveys 5 (2013), §§1, 5, provides the computational model. E. Dupont et al., *Improving the matrix multiplication exponent with modern optimization and AlphaEvolve* (2026), abstract and §1, reports $\omega < 2.371177$. The October 2026 manuscript below reports a stronger upper bound. An exponent-two claim is recorded separately with its verification limits. **Admitted: no validated resolution located.**

Status check — 2026-09-10

Rechecked Dupont et al., August 2026, and searched for exponent-two proofs and newer matrix-multiplication bounds. Its reported bound is still strictly above two, at 2.371177. No proof of exponent two or a strict lower bound above two was located. Faster finite-size identities and improved numerical optimizations of existing bounds do not decide the asymptotic equality.

Reviewed research submission — 14 September 2026

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Sections 4–8 prove the auxiliary bound $\tilde{R}(cw_2) < 3.876919161$, also relevant to AC-04. The Section 10 scalar-tree obstruction is not a tensor-rank lower bound. Neither result proves $\omega = 2$.

Status: Open. **Report · Independent AI-agent audit and limitations.** AI assistance disclosed; no external human review, formal verification or Lean checks. Original target and prior-source credit retained.

Literature update — 2026-10-06

OpenAI, *An Upper Bound of $9/4$ for the Matrix Multiplication Exponent*, manuscript dated 2 October 2026, Theorem 1.1, reports that, for every $\varepsilon > 0$, complex $n \times n$ matrix multiplication uses $O_\varepsilon(n^{9/4+\varepsilon})$ arithmetic operations, hence $\omega(\mathbb{C}) \leq 9/4 = 2.25$. The [theorem source](#) and the release's [Lean scope note](#), “Scope” are pinned to revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The scope note reports an unconditional formalization for finite division-free arithmetic programs with arbitrary positive exponent slack,

which fits the arithmetic model in this entry. This update reviewed the statement and published scope; it did not rerun Lean or independently audit the proof. The reported upper bound improves the August bound cited above but leaves $\omega = 2$ open.

Unvalidated exponent-two claim — 18 September 2026

Zhi-Wei Sun, *Tensor Degenerations for Matrix Multiplication*, ChinaXiv:202609.00167. The original [Math-ChinaXiv index](#) lists this title with date 18 September 2026. A [reproduced author abstract](#) claims that, for every positive integer n and every field $\mathbb{K}$, the border rank of the $2^n \times 2^n$ multiplication tensor is at most 2^{2n+1} , and claims the consequence $\omega(\mathbb{K}) = 2$. If established over $\mathbb{C}$, this would settle the displayed target. During this check the original abstract and PDF endpoints returned HTTP 403; the proof was inaccessible. The inspected evidence consists of the original title/date listing and the reproduced abstract. No independent validation was located, and this catalog has not checked the argument. This remains an unvalidated full-resolution claim and does not change the **Open** status.